CHEMISTRY FOR ENGINEERS

ATOM COUNT CARD



PRODUCT IDENTIFICATION

SM Bonus Refined Sugar

Net weight: 0.5 kg = 500 g

Calculation assumption: model refined table sugar as sucrose (C12H22O11). The package does not state a laboratory purity assay.

COMPOUND: SUCROSE
C12H22O11

MOLAR MASS
342.297 g/mol

AVOGADRO CONSTANT
6.02214076 x 10^23 mol^-1

01 MOLAR-MASS BUILD-UP

C: 12 x 12.011 = 144.132 g/mol
H: 22 x 1.008 = 22.176 g/mol
O: 11 x 15.999 = 175.989 g/mol
Total = 342.297 g/mol

02 LABEL BRIDGE

0.5 kg x 1000 g/kg = 500 g
500 g sugar per package
Assume 500 g is sucrose for calculation.

03 MASS TO MOLES

n = mass / molar mass
= 500 g / 342.297 g/mol
= 1.46072 mol sucrose

04 MOLES TO PARTICLES

1.46072 mol x 6.02214076 x 10^23 molecules/mol
= 8.79666 x 10^23 molecules

05 ATOMS OF ONE ELEMENT

11 oxygen atoms per sucrose molecule
8.79666 x 10^23 molecules x 11
= 9.67633 x 10^24 oxygen atoms

06 PERCENT COMPOSITION + CHECK

%C = 42.107% %H = 6.479% %O = 51.414%
Route A: 500 g x 0.514141 = 257.07 g O
Route B: 1.46072 mol x 175.989 = 257.07 g O

07 EMPIRICAL-FORMULA CHECK

100 g basis: C 42.107 g, H 6.479 g, O 51.414 g
Moles: C 3.505, H 6.427, O 3.214
Divide by 3.214: 1.091 : 2.000 : 1.000
Multiply by 11: 12 : 22 : 11
Empirical formula = C12H22O11

08 SCALE-UP + REVERSE

30 packs: 30 x 1.46072 = 43.82 mol
Mass of 1.00 mol = 342.297 g
Packages for 1 mol: 342.297 / 500 = 0.6846 pack

INTERPRETATION

The 500 g package corresponds to about 1.461 mol only under the stated sucrose assumption. The label confirms product type and net weight, not exact chemical purity.